1064 Center Dr
Gainesville, FL 32611
February 9, 2026

Dr. Kurt Schulze
University of Florida
1064 Center Dr
Gainesville, FL 32611

Dear Dr. Schulze:

The objective of this experiment is to characterize frictional losses in a piping system for cooling applications by determining the head losses, friction factors, and pipe roughness associated with two straight copper pipes of differing diameters, as well as the loss coefficient of a 90° elbow. In addition to solving the values of the above parameters, an uncertainty analysis using statistical and Root Sum Squared (RSS) methods will be carried out to determine the accuracy of results. This experiment characterizes the relationship between flow rate and pressure loss and evaluates the effect of pipe geometry and surface roughness on energy dissipation in internal flows.

The analysis was based on fundamental fluid dynamics principles such as conservation of mass and energy applied to internal pipe flow. For a steady and incompressible flow, which is assumed in this experiment, conservation of mass ensures that the mass flow rate remains constant throughout the pipe while conservation of energy ensures that the total energy (pressure, kinetic and potential) at one point equals the energy at another point minus any energy lost to friction. Fully developed flow was assumed, fluid properties were evaluated at measured bulk temperatures and gravitational acceleration was treated as a constant. Using the energy equation for horizontal pipe flow with constant diameter, head loss was found by taking the difference of the pressure terms and then dividing by the fluid's specific weight. The Reynolds number of water varied for each flow rate tested and was determined by relating properties such as density and viscosity with the fluid velocity. Next, the Darcy-Weisbach equation, typically used to calculate head loss, was rearranged to solve for the friction factor of the pipe by equating head loss, pipe dimensions, fluid velocity and the gravitational constant. Finally, the relative roughness was derived from the Colebrook equation and used to solve for absolute roughness, and entrance length was determined using the equation for turbulent flow dependent on Reynolds number and pipe diameter.

The experimental apparatus consisted of a closed-loop hydraulic circuit containing two straight copper pipes with inner diameters of 9.254 and 18.604 millimeters, and a single 90° elbow. Differential pressure transducers were positioned across each test section, and three electronic flow meters corresponding to high (35-225 LPM), medium (6-45 LPM) and low (1-9.5 LPM) flow ranges were used to measure volumetric flow rate in liters per minute with 97% accuracy. Data was acquired using a LabVIEW interface with three sampling procedures for the large pipe and one for the small pipe that determined the samples per iteration, the sampling rate in samples per second and the total run time. These sampling procedures, or runs, were tested from the maximum flowrate down to zero flowrate, with a data set being recorded at each 10% decrease interval. As the flow rate was decreased, the electronic flow meter being used had to be adjusted from high to medium to low to ensure the flow rate fell within the flow meter range for accurate results. When changing flowmeters, the valve for the new one had to be opened before the valve of the previous one was closed to avoid burning up the pipe.

The maximum flow rate and head loss for the large pipe were around 43 LPM and 14 inches, respectively, while the same metrics for the small pipe were approximately 38 LPM and 270 inches. The results demonstrated a linear relationship between head loss and flow rate squared for both pipe diameters which was consistent with theoretical expectations. Friction factor and Reynolds number for both pipes ranged from 0.0125-0.0225 and 20000 to 120000, respectively, and formed a moody chart consistent with standard copper piping and water. Based off the Reynolds number and friction factors, the flow was determined to be turbulent and a sample calculation for absolute roughness was yielded a value of 0.001484 millimeters, aligning with expectations for copper, and the entrance length for the large and small pipes at maximum flowrate were approximately 530 millimeters and 290 millimeters, respectively.

Sincerely,

Bryce Addeo | Sean Muller | Matthew Strogach

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

Table 1: This table shows experimental data results for the 18.604-millimeter diameter large pipe during run 1 which consisted of 2 samples per iteration, a sampling rate of 400 samples per second, and a 5 second run time completed in LabVIEW. Nominal flowrate was manually recorded by taking the maximum flow rate and reducing it by 10% for each new data set of run 1, while the actual flow rate was the average of the flow rate values from each data set. The uncertainty of the actual flowrate was then calculated using a statistical approach that applied a confidence interval multiplier based off the sample size of the data set, which for run 1 was roughly 108 samples. Head loss was calculated by dividing the pressure difference provided from the LabVIEW data by the specific weight of water, and the corresponding standard deviation and uncertainty was then found using an Excel function and a RSS propagation equation, respectively. Finally, the fluid temperatures were recorded by LabVIEW to document the operating conditions, and the uncertainty was calculated to account for measurement variability and instrument error. An observable trend in the data is that the head loss decreased (~14 inches to ~0 inches) as the flowrate decreased (~43 LPM to ~0 LPM). Head loss is directly proportional to the fluid velocity through the Darcy-Weisbach equation, and volumetric flow rate positively correlates to velocity and cross-sectional area. Since the pipe area was constant for the large pipe data, the decrease in flow rate can be attributed to a decrease in fluid velocity. Therefore, this decrease in fluid velocity also causes a decrease in head loss through the Darcy-Weisbach relationship. Next, the relative uncertainty of the head loss increases as flow rate decreases. While all head loss uncertainty measurements remain within the range of 0.5 to 1.7 inches, the head loss value itself is greatly reduced at lower flow rates causing the relative uncertainty to be much greater. Additionally, the zero-flow point still shows a non-zero head loss further showing that low flow data is noisier and less reliable as the pressure difference approaches the resolution limit of the differential pressure transducers. Finally, larger standard deviation of head loss was calculated at mid-range flows due to flow transitions or unsteady behavior as valves were partially closed, and temperature remained effectively constant suggesting the experiment was close to isothermal conditions.

Large pipe Nominal flowrate (LPM)	**Actual flow rate (LPM)**	**Flowrate uncertainty (± LPM)**	**Head loss Δh (inches)**	**Standard deviation of head loss (inches)**	**Total uncertainty of head loss (± inches)**	**Temperature (°C)**	**Uncertainty in temperature (°C) (std dev + instrument error)**
43.2	42.86	1.3164	14.02	2.895	0.5858	38.05	0.3027
38.9	38.62	1.1646	10.504	5.337	1.0192	38.12	0.3037
34.6	33.80	1.0214	8.238	8.17	1.5599	38.22	0.3042
30.2	30.06	0.9075	5.362	8.475	1.618	38.36	0.3025
25.9	25.51	0.7725	4.887	6.139	1.172	38.39	0.3038
21.6	21.49	0.6547	3.3799	6.437	1.2289	38.501	0.3035
17.3	17.42	0.5283	2.2789	5.8933	1.125	38.58	0.3033
13.0	12.92	0.3911	1.3108	3.5407	1.0893	38.67	0.3013
8.6	8.48	0.2559	0.3452	4.5665	0.8717	38.49	0.3037
4.3	4.24	0.1324	0.8143	4.0947	0.7817	38.61	0.3005
0.0	-0.02	0.00327	0.1073	4.1404	0.7904	38.53	0.3047

Table 2: This table shows experimental data results for the large pipe during run 2 which consisted of 200 samples per iteration, a sampling rate of 400 samples per second, and a 5 second run time completed in LabVIEW. All values were recorded or calculated with the same method as Table 1, with the only difference being the sample size of the data and therefore the calculation of statistical uncertainty. The data sets for run 2 featured 11 samples, and therefore a greater confidence interval multiplier was applied to the standard deviation, and that product was then divided by a smaller number (square root of sample size) to solve statistical uncertainty for flowrate, pressure difference and temperature. The head loss and flow rate values, and the trend between them, remained similar to that of run 1, suggesting that the different sampling procedure had little impact on those metrics. One notable difference is that the standard deviation of head loss for run 2 was much lower than that of run 1. This is due to the differences in the sampling procedures that resulted in the run 1 data sets having 108 samples and the run 2 data sets having only 11 samples. With nearly ten times more samples in run 1 than run 2, there were more opportunities for inaccurate data to be recorded thus contributing to the higher standard deviation. Despite the differences in samples, the temperature still remained roughly constant in run 2, further suggesting an isothermal experimental condition.

Large pipe Nominal flowrate (LPM)	**Actual flow rate (LPM)**	**Flowrate uncertainty (± LPM)**	**Head loss Δh (inches)**	**Standard deviation of head loss (inches)**	**Total uncertainty of head loss (± inches)**	**Temperature (°C)**	**Uncertainty in temperature (°C) (std dev + instrument error)**
43.2	42.76	1.385	13.766	0.0554	0.0529	38.06	0.3014
38.9	38.61	1.2373	10.855	0.1958	0.1333	38.16	0.3015
34.6	33.83	1.0566	8.315	0.1831	0.1237	38.26	0.3020
30.2	29.98	0.9379	6.771	0.2339	0.1564	38.37	0.3014
25.9	25.41	0.8068	5.218	0.1655	0.1108	38.41	0.3003
21.6	21.55	0.6693	3.808	0.0869	0.0586	38.53	0.3134
17.3	17.42	0.5424	2.574	0.0720	0.0482	38.61	0.3004
13.0	12.89	0.4215	1.452	0.1038	0.0689	38.69	0.3135
8.6	8.50	0.2639	0.4233	0.0583	0.0387	38.66	0.3144
4.3	4.22	0.1476	0.0922	0.0982	0.0652	38.57	0.3013
0.0	-0.04	-0.00115	0.0169	0.0730	0.0485	38.46	0.3083

Table 3: This table shows experimental data results for the large pipe during run 3 which consisted of 200 samples per iteration, a sampling rate of 400 samples per second, and a 10 second run time completed in LabVIEW. All values were recorded or calculated with the same method as the previous tables, but likewise to table 2, table 3 also featured data sets with a different sample size from table 1 which impacted the calculation of statistical uncertainty. The data sets for run 3 featured 20 samples, placing them in between the sample sizes for run 1 and run 2 data. Therefore, a multiplier greater than that used in run 1 but less than that used in run 2 was applied to the standard deviation, and the product was divided by a smaller number than run 1 but a greater number than run 2 to solve statistical uncertainty. All the same trends as table 2 can be observed for table 3 including the reduced standard deviation of head loss from run 1 since run 3 featured a sample size much closer in value to run 2.

Large pipe Nominal flowrate (LPM)	**Actual flow rate (LPM)**	**Flowrate uncertainty (± LPM)**	**Head loss Δh (inches)**	**Standard deviation of head loss (inches)**	**Total uncertainty of head loss (± inches)**	**Temperature (°C)**	**Uncertainty in temperature (°C) (std dev + instrument error)**
43.2	42.88	1.3695	13.798	0.0541	0.0458	38.06	0.3011
38.9	38.64	1.1846	10.850	0.1513	0.0767	38.16	0.3059
34.6	33.86	1.0318	8.2898	0.1805	0.0873	38.28	0.3096
30.2	30.02	0.9292	6.6468	0.1666	0.0799	38.37	0.3010
25.9	25.44	0.7847	5.1837	0.1294	0.0620	38.46	0.3044
21.6	21.54	0.6647	3.7753	0.1004	0.0479	38.57	0.3011
17.3	17.62	0.5425	2.6456	0.1134	0.0534	38.65	0.3087
13.0	12.90	0.4065	1.4618	0.1069	0.0500	38.73	0.3097
8.6	8.50	0.2585	0.4090	0.0893	0.0417	38.69	0.3106
4.3	4.23	0.1379	0.1206	0.1146	0.0535	38.58	0.3079
0.0	-0.04	-0.00116	0.0100	0.0801	0.0374	38.30	0.3179

Table 4: This table shows experimental data results for the 9.254-millimeter diameter small pipe during its singular run which consisted of 200 samples per iteration, a sampling rate of 400 samples per second, and a 10 second run time completed in LabVIEW. Since the sampling procedure consisted of the same metrics as run 3 for the large pipe, all the data sets had the same sample size as those analyzed in table 3. Therefore, the data and results below were recorded or calculated with the exact same methods as table 3. However, notable differences from the large pipe data include a lower maximum flow rate (~38 LPM versus ~43 LPM) and much larger head loss values (~270 inches versus ~14 inches at maximum flow rate). The maximum flow rate in the small pipe is lower than that of the large pipe because the hydraulic resistance of the pipe increases as the diameter decreases as governed by the Darcy-Weisbach equation. Similarly, the head loss is much greater for the small pipe than the large pipe because the flow rate divided by the smaller area results in a greater fluid velocity, and head loss is proportional to velocity squared through the Darcy-Weisbach relation. Since the relationship of velocity to head loss is quadratic, an increase in velocity results in an even greater increase in head loss as reflected in the small pipe data.

Small pipe Nominal flowrate (LPM)	**Actual flow rate (LPM)**	**Flowrate uncertainty (± LPM)**	**Head loss Δh (inches)**	**Standard deviation of head loss (inches)**	**Total uncertainty of head loss (± inches)**	**Temperature (°C)**	**Uncertainty in temperature (°C) (std dev + instrument error)**
38.2	37.95	1.1884	269.7	0.5066	0.7837	39.18	0.3073
34.4	33.97	1.0367	210.7	0.5686	0.6410	39.35	0.3111
30.6	30.07	0.9372	166.95	0.5509	0.5291	39.38	0.3029
26.7	26.73	0.821	135.51	0.3549	0.4103	39.46	0.3119
22.9	22.76	0.6949	101.02	0.2887	0.3105	39.47	0.3068
19.1	19.28	0.5889	74.51	0.2392	0.2346	39.48	0.3012
15.3	15.23	0.4716	48.12	0.1751	0.1563	39.48	0.3010
11.5	11.53	0.3582	28.25	0.1168	0.0953	39.49	0.3062
7.6	7.58	0.2338	6.55	0.0639	0.0349	39.38	0.3010
3.8	3.85	0.1182	0.1770	0.0728	0.0339	39.26	0.3087
0.0	-0.03	0.0005	-2.981	0.0629	0.0305	39.13	0.3090

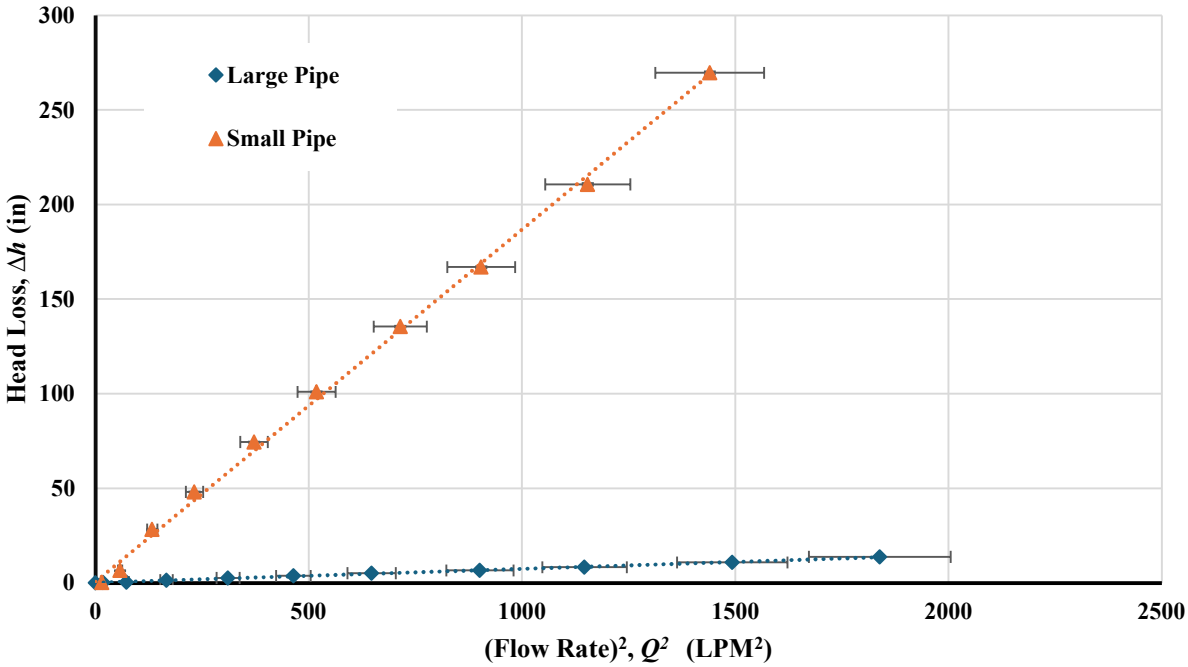

Figure 1: Flow rate data was taken using three electronic flow meters for high, medium and low flow rates. Head loss was calculated based off pressure differential measurements from LabVIEW and the specific weight of water. Run 3 data for the large pipe was used for this graph because it featured the same sampling procedure and therefore sample size as the run for the small pipe. The units for flowrate were left in liters-per-minute (LPM) despite the head loss being in inches because multiplying each value by a constant conversion factor would have no impact on the shape of the graph which is the important factor to observe in this case. A linearly increasing relationship can be observed between head loss and flow rate squared because frictional head loss in pipe flow is proportional to velocity squared and velocity is directly proportional to flow rate via the Darcy-Weisbach equation. Error bars are included at each point to visually show the uncertainty for flow rate squared and head loss. The horizontal error bars correspond to the uncertainty of flow rate squared which was calculated using an RSS propagation equation. The negative and positive sides of the error bars are equal in length because all uncertainties are calculated as plus or minus a single value. Finally, the vertical error bars corresponding to the head loss uncertainty cannot be seen on the graph because the values are too small for the bars to emerge past the markers.

Sample Calculation of Head Loss:

The following sample calculation solves the head loss, Δh, of the first data set in run 1 using the equation below where ΔP is the difference in pressure record by LabVIEW, ρ is the density of water and g is the gravitational constant.

$$\Delta h = \frac{\Delta P}{\rho g} \quad p.7\ Schulze\ [1] \tag{1}$$

Since the units of the pressure difference recorded by LabVIEW were in psi (lbf/in^2), and head loss needs to be solved in inches, U.S. customary units will be used for all other variables. However, this raises an issue when using density because the pound-mass (lbm) unit does not cancel out with the pound-force (lbf) unit from the pressure differential. Therefore, to align the units, the specific weight of water, γ, will be substituted in (1) using (2) below.

$$\gamma = \rho g \quad Eqn.2.4, p.17\ Fox\ \&\ McDonald\ [2] \tag{2}$$

The resulting equation that was used in the calculation is shown below in (3).

$$\Delta h = \frac{\Delta P}{\gamma} \tag{3}$$

The values for the pressure differential and the specific weight of water are presented below. The pressure value was provided by LabVIEW and the specific weight value for the average water temperature of the data set equal to 38.05 °C was used [3].

$$\Delta P = 0.5061\ \frac{lbf}{in^2}$$

$$\gamma = 0.0361\ \frac{lbf}{in^3}$$

$$\Delta h = \frac{\Delta P}{\gamma} = \frac{0.5061\ \frac{\cancel{lbf}}{\cancel{in^2}}}{0.0361\ \frac{\cancel{lbf}}{in^{\cancel{3}}}}$$

$$\therefore \Delta h = 14.02\ in$$

Sample Calculation of Head Loss Uncertainty:

The uncertainty of the head loss was found using a Root Sum Squared (RSS) propagation equation that involved the uncertainties of all other variables involved. The first data set of run 1 for the large pipe is used again for this sample calculation. The RSS equation for the uncertainty of head loss, $\mu_{\Delta h}$, is shown below where $\mu_{\Delta P}$ is the statistical uncertainty of the pressure differential, μ_γ is the uncertainty of the specific weight of water, $\frac{\partial \Delta h}{\partial \Delta P}$ is the partial derivative of head loss with respect to the pressure differential and $\frac{\partial \Delta h}{\partial \gamma}$ is the partial derivative of head loss with respect to the specific weight of water.

$$\mu_{\Delta h} = \sqrt{(\mu_{\Delta P}\frac{\delta \Delta h}{\delta \Delta P})^2 + (\mu_\gamma \frac{\delta \Delta h}{\delta \gamma})^2} \tag{4}$$

The methods of calculating the values of the above terms are shown below. Note that the calculation of the pressure differential uncertainty depended on the standard deviation of the pressure differential, $\sigma_{\Delta P}$, which was found using an Excel function, a 95% confidence interval multiplier, t, and the sample size, N.

$$\mu_{\Delta P} = \frac{t\sigma_{\Delta P}}{\sqrt{N}} = \frac{1.984\sigma_{\Delta P}}{\sqrt{108}} = \frac{1.984(0.5061\ \frac{lbf}{in^2})}{\sqrt{108}} = 0.01995\ \frac{lbf}{in^2}$$

$$\mu_\gamma = 0.0005\frac{lbf}{in^3}\ [3]$$

$$\frac{\delta \Delta h}{\delta \Delta P} = \frac{\delta(\frac{\Delta P}{\gamma})}{\delta \Delta P} = \frac{1}{\gamma} = \frac{1}{0.0361} = 27.7008\ \frac{in^3}{lbf}$$

$$\frac{\delta \Delta h}{\delta \gamma} = \frac{\delta(\frac{\Delta P}{\gamma})}{\delta \gamma} = -\frac{\Delta P}{\gamma^2} = -\frac{0.5061\ \frac{lbf}{in^2}}{\left(0.0361\ \frac{lbf}{in^3}\right)^2} = -\frac{0.5061\ \frac{\cancel{lbf}}{\cancel{in^2}}}{0.0013\ \frac{lbf^{\cancel{2}}}{in^{\cancel{6}}}} = -388.3118\ \frac{in^4}{lbf}$$

The above values can now be plugged into their respective variables to solve for the uncertainty of head loss as shown below.

$$\mu_{\Delta h} = \sqrt{(\mu_{\Delta P}\frac{\delta \Delta h}{\delta \Delta P})^2 + (\mu_\gamma \frac{\delta \Delta h}{\delta \gamma})^2} = \sqrt{((0.01995\ \frac{\cancel{lbf}}{\cancel{in^2}})(27.7008\ \frac{in^{\cancel{3}}}{\cancel{lbf}}))^2 + (\left(0.0005\frac{\cancel{lbf}}{\cancel{in^3}}\right)\left(-388.3118\ \frac{in^4}{\cancel{lbf}}\right))^2}$$

$$\mu_{\Delta h} = \sqrt{(0.5526\ in)^2 + (-0.1942\ in)^2}$$

$$\therefore \mu_{\Delta h} = 0.5858\ in$$

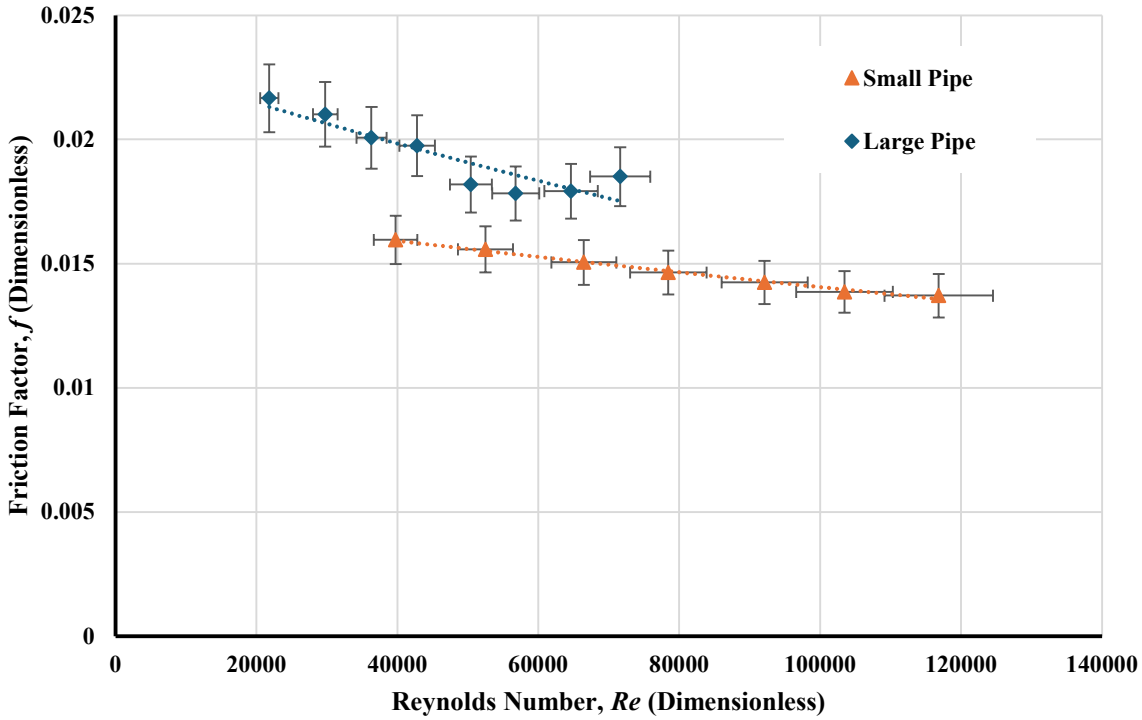

Figure 2: For both the large pipe and the small pipe, the Reynolds number and friction factor were determined to characterize flow behavior and energy losses from friction in the pipes. The Reynolds number was calculated based off fluid density, velocity, the hydraulic diameter of the pipe, and fluid viscosity. The friction factor was calculated based off gravity, head loss, hydraulic diameter, pipe length, and fluid velocity. The data from run 3 for the large pipe was used to match the sampling procedure of the small pipe run consisting of 200 samples per iteration, a sampling rate of 400 samples per second, and a 10 second run time. Both variables are dimensionless, so when calculating them all units were converted to an SI format, although the units would not affect the outcome if they were consistent. Both pipes experienced turbulent flow with all Reynolds numbers being greater than 20,000. As Reynolds number increases, the friction factor decreases, which makes sense. This is because in smooth pipes, as the Reynolds Number increases, viscous effects become less dominant compared to effects from inertia, therefore friction loss becomes smaller. The small pipe data portrays lower friction factor values than the large pipe data which is caused by the difference in diameter and thus cross-sectional area. Overall, the data confirms a strong turbulent behavior. The error bars in this graph came from the performed RSS propagation for both variables and increase at a comparable rate to the values.

Sample Calculation of Reynolds Number:

The following calculation solves for the Reynolds number, which describes the flow of fluid through a pipe relating its internal forces to viscous properties. This sample calculation was based off the third data set of the small pipe run where ρ is the fluid density, V is the fluid velocity, D_h is the hydraulic diameter, and μ is the dynamic viscosity.

$$Re = \frac{\rho V D_h}{\mu} \quad p.10\ Schulze\ [1] \tag{5}$$

Since Reynolds number is a dimensionless value, all units were made consistent with one another to fully cancel out. The density was found based on the temperature the fluid was traveling at when density was recorded [3]. Velocity was determined based on the actual volumetric flowrate and area to be finalized in units of $\frac{m}{s}$. The hydraulic diameter was determined from caliper measurements and converted to m [1]. Lastly, the viscosity was found based on the average fluid temperature of the data set [4].

$$Q = 22.76\ \cancel{LPM} * \frac{10^{-3} m^3}{1\ \cancel{L}} * \frac{1\ \cancel{min}}{60\ s} = 0.0003793\ \frac{m^3}{s}$$

$$D_h = 9.254\ \cancel{mm} * \frac{1\ m}{1000\ \cancel{mm}} = 0.009254\ m$$

$$V = \frac{Q}{A} = \frac{0.00\ \frac{m^{\cancel{3}}}{s}}{\pi * \frac{(0.009254\ \cancel{m})^{\cancel{2}}}{4}} = 5.6399\ \frac{m}{s}$$

$$= \frac{992.45\ \frac{\cancel{kg}}{\cancel{m^3}} * 5.6399\ \frac{\cancel{m}}{\cancel{s}} * 0.009254\ \cancel{m}}{0.0006604\ \frac{\cancel{kg}}{\cancel{m * s}}}$$

$$\therefore Re = 78433.7833$$

Sample Calculation of Reynolds Number Uncertainty:

The uncertainty in the Reynolds number was found using an RSS propagation equation where the partial derivative of the Reynolds number was taken with respect to each variable. However, the uncertainty of each variable in the Reynolds number equation was first found. The density uncertainty was determined to be 0.05% of the actual measured value [5]. The velocity uncertainty was determined through RSS propagation as well, since velocity was calculated using (6) based off the measured flow rate and pipe diameter. The hydraulic diameter uncertainty was found by taking half of the caliper resolution, which was 0.0005 in, or 0.0000127 m. Lastly, the viscosity uncertainty was found by determining 5% of the measured value [6].

$$\mu_\rho = 992.45\ \frac{kg}{m^3} * 0.0005 = 0.4962\ \frac{kg}{m^3}$$

$$V = \frac{Q}{A} = \frac{4Q}{\pi D_h^2}\ Eqn.4.14, p.78\ \text{Fox \& McDonald [2]} \tag{6}$$

$$\mu_V = \sqrt{(\mu_Q \frac{\delta V}{\delta Q})^2 + (\mu_{D_h} \frac{\delta V}{\delta D_h})^2} \tag{7}$$

$$\frac{\delta V}{\delta Q} = \frac{4}{\pi * D_h^2} = \frac{4}{\pi * 0.009254\ m^2} = 14867.9512\ \frac{1}{m^2}$$

$$\frac{\delta V}{\delta D} = -\frac{8 * Q}{\pi * D_h^3} = -\frac{8 * 0.0003793\ \frac{\cancel{m^3}}{s}}{\pi *\ 0.009254\ \cancel{m^3}} = -1218.9127\ \frac{1}{s}$$

$$\mu_V = \sqrt{(\mu_Q * \frac{4}{\pi * D_h^2})^2 + (\mu_{D_h} * -\frac{8 * Q}{\pi * D_h^3})^2}$$

$$\mu_V = \sqrt{(0.00001158\ \frac{m^{\cancel{3}}}{s} *\ 14867.9513\ \frac{1}{\cancel{m^2}})^2 + (0.0000127\ m *\ -1218.9127\ \frac{1}{s})^2} = 0.1729\ \frac{m}{s}$$

$$\mu_\mu = 0.0006604\ \frac{kg}{m * s} * 0.05 = 0.00003302\ \frac{kg}{m * s}$$

Each partial derivative could then be solved and was multiplied by the uncertainty of the variable it was taken with respect to. Those products were then squared, and the square root of the sum of all the terms was calculated as Reynolds number uncertainty.

$$\mu_{Re} = \sqrt{(\mu_\rho \frac{\delta Re}{\delta \rho})^2 + (\mu_V \frac{\delta Re}{\delta V})^2 + (\mu_{D_h} \frac{\delta Re}{\delta D_h})^2 + (\mu_\mu \frac{\delta Re}{\delta \mu})^2} \tag{8}$$

$$\frac{\delta Re}{\delta \rho} = \frac{V * D_h}{\mu} = \frac{5.6399091\ \frac{m}{\cancel{s}} * 0.009254\ m}{0.0006604\ \frac{kg}{m\cancel{s}}} = 79.03046\ \frac{m^3}{kg}$$

$$\frac{\delta Re}{\delta V} = \frac{\rho * D_h}{\mu} = \frac{992.45\,\frac{\cancel{kg}}{\cancel{m^3}} * 0.009254\,\cancel{m}}{0.0006604\,\frac{\cancel{kg}}{\cancel{m}s}} = 13906.924\ \frac{s}{m}$$

$$\frac{\delta Re}{\delta D_h} = \frac{\rho * V}{\mu} = \frac{992.45\,\frac{\cancel{kg}}{\cancel{m^3}} * 5.6399\,\frac{\cancel{m}}{\cancel{s}}}{0.0006604\,\frac{\cancel{kg}}{\cancel{m}\cancel{s}}} = 8475662.8\ \frac{1}{m}$$

$$\frac{\delta Re}{\delta \mu} = -\frac{\rho * D_h * V}{\mu^2} = -\frac{992.45\,\frac{\cancel{kg}}{\cancel{m^3}} * 0.009254\,\cancel{m} * 5.6399\,\frac{\cancel{m}}{\cancel{s}}}{\left(0.0006604\,\frac{kg}{ms}\right)^{\cancel{2}}} = -118767085.6\ \frac{m*s}{kg}$$

$$\mu_{Re} = \sqrt{(\mu_\rho(\frac{V * D_h}{\mu})^2 + (\mu_V * \frac{\rho * D_h}{\mu})^2 + (\mu_{D_h} * \frac{\rho * V}{\mu})^2 + (\mu_\mu * -\frac{\rho * D_h * V}{\mu^2})^2}$$

$$= \sqrt{\begin{array}{c}\left(0.49623\,\frac{\cancel{kg}}{\cancel{m^3}} * 79.03046\,\frac{\cancel{m^3}}{\cancel{kg}}\right)^2 + \left(0.1729\,\frac{\cancel{m}}{\cancel{s}} * 13906.924\,\frac{\cancel{s}}{\cancel{m}}\right)^2 \\ + \left(0.0000127\,\cancel{m} * 8475662.8\,\frac{\cancel{1}}{\cancel{m}}\right)^2 + \left(0.00003302\,\frac{\cancel{kg}}{\cancel{m * s}} * -118767085.6\,\frac{\cancel{m * s}}{\cancel{kg}}\right)^2\end{array}}$$

$$\therefore \mu_{Re} = 4601.4956$$

Sample Calculation of Friction Factor

The following calculation was done to determine the friction factor, which is another dimensionless variable that represents the resistance to fluid flow within a pipe. The calculation is for the third data set of the small pipe run. To determine the friction factor, the Darcy Weisbach equation was used, which is dependent on gravity, head loss, hydraulic diameter, length of the pipe, and velocity.

$$f = \frac{2g\Delta h D_h}{LV^2} \quad Eqn.1, EngineerExcel\ [7] \tag{9}$$

All variables were converted to SI units to be consistent with one another and ensure unit conversion errors had no effect on the friction factor. The standard gravity value of 9.81 $\frac{m}{s^2}$ was used. The length was provided to be 1000 millimeters which was converted to meters.

$$L = 1000\ \cancel{mm} * \left(\frac{1\ m}{1000\ \cancel{mm}}\right) = 1\ m$$

$$g = 9.81\frac{m}{s^2}$$

$$\Delta h = \frac{\Delta P}{\gamma} = 2.5659\ m$$

$$D_h = 0.009254\ m$$

$$V = \frac{4Q}{\pi D_h^2} = 5.6399\frac{m}{s}$$

$$f = \frac{2g\Delta h D_h}{LV^2}$$

$$= \frac{2*9.81\frac{m}{s^2}*2.56 \qquad *0.009254\ m}{1\ m*(5.6399\frac{m}{s})^2}$$

$$\therefore f = 0.01465$$

Sample Calculation of Friction Factor Uncertainty:

The uncertainty of the friction factor was found through RSS propagation based on the uncertainty of gravity, head loss, hydraulic diameter, pipe length, and velocity as well as all the partial derivatives with respect to each variable. The uncertainty of gravity was 0 $\frac{m}{s^2}$ since it is a constant value. The uncertainty of pipe length was set to 0.5 mm or 0.0005 m because the pipe length was provided and not measured during experimental performance [8]. The diameter uncertainty was set to half the caliper resolution of 0.0000127 m. The head loss and velocity uncertainties were found through RSS propagation as previously shown.

$$\mu_f = \sqrt{(\mu_g \frac{\delta f}{\delta g})^2 + (\mu_{\Delta h} \frac{\delta f}{\delta \Delta h})^2 + (\mu_{D_h} \frac{\delta f}{\delta D_h})^2 + (\mu_L \frac{\delta f}{\delta L})^2 + + (\mu_V \frac{\delta f}{\delta V})^2} \tag{10}$$

$$\mu_{\Delta h} = 0.000005175\ m$$

$$\mu_V = 0.1729\ \frac{m}{s}$$

$$\frac{\delta f}{\delta g} = \frac{2 * \Delta h * D_h}{L * V^2} = \frac{2 * 2.5659\ \cancel{m} * 0.009254\ \cancel{m}}{1\ \cancel{m} * (5.6399 \frac{m}{s})^2} = 0.0014929\ \frac{s^2}{m}$$

$$\frac{\delta f}{\delta \Delta h} = \frac{2 * g * D_h}{L * V^2} = \frac{2 * 9.81\ \frac{\cancel{m}}{\cancel{s^2}} * 0.009254\ \cancel{m}}{1\ \cancel{m} * (5.6399\ \frac{m}{s})^{\cancel{2}}} = 0.005708\ \frac{1}{m}$$

$$\frac{\delta f}{\delta D_h} = \frac{2 * \Delta h * g}{L * V^2} = \frac{2 * 2.5659\ \cancel{m} * 9.81\ \frac{\cancel{m}}{\cancel{s^2}}}{1\ m * (5.6399\ \frac{\cancel{m}}{\cancel{s}})^{\cancel{2}}} = 1.5827\ \frac{1}{m}$$

$$\frac{\delta f}{\delta L} = -\frac{2 * \Delta h * D_h * g}{L^2 * V^2} = -\frac{2 * 2.5659\ \cancel{m} * 0.009254\ \cancel{m} * 9.81\ \frac{\cancel{m}}{\cancel{s^2}}}{(1\ m)^{\cancel{2}} * \left(5.6399\ \frac{\cancel{m}}{\cancel{s}}\right)^{\cancel{2}}} = -0.01465\ \frac{1}{m}$$

$$\frac{\delta f}{\delta V} = -\frac{4 * \Delta h * D_h * g}{L * V^3} = -\frac{4 * 2.5659\cancel{m} * 0.009254\ \cancel{m} * 9.81\ \frac{\cancel{m}}{\cancel{s^2}}}{1\ m * \left(5.6399\ \frac{\cancel{m}}{s}\right)^{\cancel{3}}} = -0.005194\ \frac{s}{m}$$

$$\mu_f = \sqrt{\left(\mu_g * \frac{2 * \Delta h * D_h}{L * V^2}\right)^2 + \left(\mu_{\Delta h} * \frac{2 * g * D_h}{L * V^2}\right)^2 + \left(\mu_{D_h} * \frac{2 * \Delta h * g}{L * V^2}\right)^2 + (\mu_L * -\frac{2 * \Delta h * D_h * g}{L^2 * V^2})^2 + (\mu_V * \frac{4 * \Delta h * D_h * g}{L * V^3})^2}$$

$$\mu_f = \sqrt{\left(0\ \frac{\cancel{m^2}}{\cancel{s}} * 0.001493\ \frac{\cancel{s^2}}{\cancel{m}}\right)^2 + \left(0.000005175\ \cancel{m} * 0.005708\ \frac{\cancel{1}}{\cancel{m}}\right)^2 + \left(0.0000127\ \cancel{m} * 1.5827\ \frac{\cancel{1}}{\cancel{m}}\right)^2 + (0\ \cancel{m} * -0.01465\ \frac{\cancel{1}}{\cancel{m}})^2 + + (0.1729\ \frac{\cancel{m}}{\cancel{s}} * -0.005194\ \frac{\cancel{s}}{\cancel{m}})^2}$$

$$\therefore \mu_f = 0.0008982$$

Table 5: This table shows experimental data results for the 90° elbow during run 3 for the large pipe. The same flow rate measurements as table 3 were recorded, however, the head loss values were now calculated using the pressure differential of the elbow rather than the straight large pipe. Whereas the maximum head loss for the large pipe was approximately 14 inches for run 3, the maximum head loss for the elbow is roughly 21 inches, exhibiting a 50% increase. This is because the 90° elbow introduces additional energy losses that do not exist in straight pipe flow, even if the diameters are the same. The standard deviation and head loss were calculated in the same way as Table 3, but the values for the elbow were lower, suggesting that the pressure differential measurements were more consistently accurate. Finally, unlike the previous tables, a loss coefficient and corresponding uncertainty were calculated for each data set of the run. The loss coefficient is a dimensionless measure of how much mechanical energy is dissipated by the elbow due to its geometry, independent of pipe length. It was calculated using the head loss, fluid velocity, and the gravitational constant. No clear trend can be observed for the loss coefficient, other than the fact it stayed consistent around 1.4-1.5 for most data sets, but then became highly inaccurate for low flow conditions. This loss of accuracy at low flow rates can be attributed to the true elbow head loss approaching the resolution limit of the pressure transducers causing the data to be overtaken by noise.

Elbow Nominal flowrate (LPM)	Actual flow rate (LPM)	Flowrate uncertainty (± LPM)	Head loss Δh (inches)	Standard deviation of head loss (inches)	Total uncertainty of head loss (± inches)	Loss Coefficient	Loss Coefficient Uncertainty
43.2	42.88	1.3695	21.2644	0.0047	0.002194	1.5322	0.09743
38.9	33.64	1.1846	16.2432	0.00718	0.003348	1.9017	0.1336
34.6	33.86	1.0317	12.2522	0.00585	0.002731	1.4159	0.08605
30.2	30.02	0.9292	9.5940	0.00592	0.002763	1.4105	0.08727
25.9	25.44	0.7847	7.1102	0.00534	0.002490	1.4556	0.08994
21.6	21.53	0.6647	5.0616	0.00601	0.002802	1.4468	0.09093
17.3	17.62	0.5425	3.3019	0.00488	0.002277	1.4091	0.08952
13.0	12.9	0.4065	1.6234	0.00455	0.002121	1.2925	0.09243
8.6	8.5	0.2585	0.2150	0.00355	0.001654	0.3942	0.08716
4.3	4.23	0.1379	-0.1431	0.00394	0.001838	-1.0594	0.3824
0.0	-0.04	-0.001158	-0.2918	0.00512	0.002387	-24163.4	8060.2351

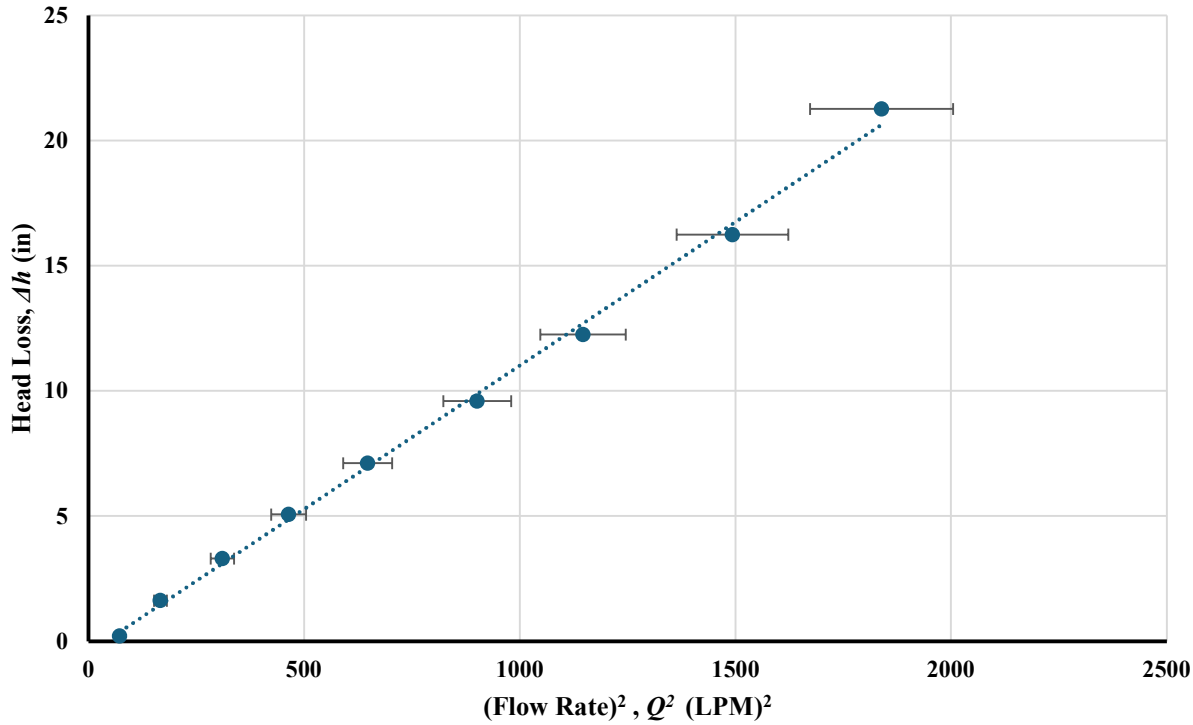

Figure 3: This graph portrays the relationship between $(Flow\ Rate)^2\ Q^2$, in units of LPM^2 and head loss of the elbow, *Δh,* in units of inches. This data was taken from run 3 of the large pipe. The head loss here was specifically calculated for the 90-degree elbow by using the pressure differential measured at the elbow. Overall, the data portrays a strong linear relationship between the two variables, consistent with the minor head loss equation. In this graph, the loss coefficient represents a proportion between lost energy across the 90-degree elbow and kinetic energy of the fluid flow. The slope of the linear trendline explains how much head loss increases as the square of flow rate also increases. The steeper the slope is, the larger the loss coefficient which means there is a greater energy loss through the elbow.

Sample Calculation of Loss Coefficient:

The loss coefficient was determined for a 90-degree elbow at 100% flow rate which related head loss across the fitting to the velocity of the flow. Volumetric flowrate, Q, was determined to calculate the velocity through the pipe followed by finding the corresponding head loss, Δh, which was obtained from the pressure differential in the elbow and specific weight of water. Loss coefficient also depends on gravity, g, which is a constant equal to 386 $\frac{in}{s^2}$. The loss coefficient was found by rearranging the equation for minor head loss.

$$\Delta h = \frac{K_L V^2}{2g} \; p.5 \; Schulze \; [8] \tag{11}$$

$$\Delta h = \frac{K_L V^2}{2g}$$

$$K_L = \frac{2g\Delta h}{V^2}$$

$$K_L = \frac{2 * 386 \frac{\cancel{in}}{\cancel{s^2}} * 7.1102 \; \cancel{in}}{(61.4087 \frac{\cancel{in}}{\cancel{s}})^2}$$

$$\therefore K_L = 1.4556$$

Loss Coefficient Uncertainty Sample Calculation

The uncertainty of the loss coefficient was determined through an RSS propagation. The individual uncertainties for gravity, head loss, and velocity were needed to complete this. The uncertainty of gravity is $0\ \frac{in}{s^2}$ due to it being a constant value. The uncertainty of head loss in the elbow was found using (4) which was previously worked out. The velocity uncertainty was also found using RSS propagation which would come from contributions of the volumetric flowrate and diameter uncertainty.

$$\mu_g = 0\ \frac{in}{s^2}$$

$$\mu_{\Delta h} = \sqrt{(\mu_{\Delta P}\frac{\delta \Delta h}{\delta \Delta P})^2 + (\mu_\gamma \frac{\delta \Delta h}{\delta \gamma})^2} = 0.0717\ in$$

$$\mu_V = \sqrt{(\mu_Q \frac{\delta V}{\delta Q})^2 + (\mu_{D_h}\frac{\delta V}{\delta D_h})^2} \tag{12}$$

$$\frac{\delta V}{\delta Q} = \frac{4}{\pi * D_h^2} = \frac{4}{\pi * 0.7324\ in^2} = 2.3734\ \frac{1}{in^2}$$

$$\frac{\delta V}{\delta D} = -\frac{8 * Q}{\pi * D_h^3} = -\frac{8 * 25.8741\ \frac{\cancel{in^3}}{s}}{\pi *\ 0.7324\ \cancel{in^3}} = -167.6822\ \frac{1}{s}$$

$$\mu_V = \sqrt{(\mu_Q * \frac{4}{\pi * D_h^2})^2 + (\mu_{D_h} * -\frac{8 * Q}{\pi * D_h^3})^2}$$

$$\mu_V = \sqrt{(0.7981\frac{in^3}{s} *\ 2.3734\ \frac{1}{\cancel{in^2}})^2 + (0.0005\ in *\ -167.6822\ \frac{1}{s})^2} = 1.8718\ \frac{in}{s}$$

Once all the variable uncertainties were found, the partial derivatives in the RSS equation for loss coefficient could be solved leading to a final loss coefficient uncertainty value.

$$\mu_{K_L} = \sqrt{(\mu_g \frac{\delta K_L}{\delta g})^2 + (\mu_{h_L}\frac{\delta K_L}{\delta \Delta h})^2 + (\mu_V \frac{\delta K_L}{\delta V})^2} \tag{13}$$

$$\frac{\delta K_L}{\delta g} = \frac{2 * \Delta h}{V^2} = \frac{2 * 7.1102\ \cancel{in}}{(61.4086\ \frac{in}{s})^2} = 0.003771\ \frac{s^2}{in}$$

$$\frac{\delta K_L}{\delta \Delta h} = \frac{2 * g}{V^2} = \frac{2 * 386\ \frac{\cancel{in}}{\cancel{s^2}}}{(61.4087\ \frac{in}{\cancel{s}})^2} = 0.20472\ \frac{1}{in}$$

$$\frac{\delta K_L}{\delta V} = -\frac{4 * g * \Delta h}{V^3} = -\frac{2 * 386\ \frac{\cancel{in}}{\cancel{s^2}} *\ 7.1102\ \cancel{in}}{(61.4087\ \frac{in}{s})^3} = -0.04741\ \frac{s}{in}$$

$$\mu_{K_L=} \sqrt{(0\ \frac{\cancel{in}}{\cancel{s^2}} *\ 0.003771\ \frac{\cancel{s^2}}{\cancel{in}})^2 + (0.002490\ \cancel{in} *\ 0.2047\ \frac{\cancel{1}}{\cancel{in}})^2 + (1.8718\ \frac{\cancel{in}}{\cancel{s}} * -0.04741\ \frac{\cancel{s}}{\cancel{in}})^2}$$

$$\therefore\ \mu_{K_L=}\ 0.08873$$

Sample Roughness Factor Calculation

The roughness factor of the large pipe was determined by using the Colebrook equation for friction factors in turbulent flow that is present on a Moody chart. It is based on the Reynolds Number, *Re*, and friction factor, *f*, of the turbulent fluid flow through the pipe. When fluid is flowing through a pipe, the roughness factor, or the relative roughness, explains how any irregularities in the pipe surface affect head loss from friction. The higher the roughness factor is, the less smooth the pipe is which leads to slower fluid velocity and more energy loss. The roughness factor is a dimensionless variable and represents what the absolute roughness actually is, but scaled by pipe diameter. Relative roughness tends to decrease as the Reynolds number decreases. A sample of the roughness factor was calculated for the eighth data set for run 3 of the large pipe.

$$\frac{\varepsilon}{D} = 3.7\left[10^{-\frac{1}{2*\sqrt{f}}} - \frac{2.51}{Re * \sqrt{f}}\right] \; Eqn.2, Moody \; [9] \tag{14}$$

$$\frac{\varepsilon}{D} = 3.7\left[10^{-\frac{1}{2*\sqrt{0.02166}}} - \frac{2.51}{21811.8798 * \sqrt{0.02166}}\right] = 0.0001421$$

The calculated roughness factor was small indicating that fluid flows smoothly and there is minimal pressure loss. In order to determine the absolute roughness of the pipe, the relative roughness values for all data sets in run 3 were averaged. The absolute roughness was then solved by multiplying the average relative roughness by the large pipe diameter.

$$\frac{\varepsilon}{D_{average}} = \frac{0.007977}{10} = 0.00007977$$

$$\frac{\varepsilon_{absolute}}{D} = 0.00007977$$

$$\frac{\varepsilon_{absolute}}{18.604\ mm} = 0.00007977$$

$$\varepsilon_{absolute} = 0.001484\ mm$$

The absolute roughness of the pipe indicates smooth flow and a very strong surface finish. This verifies that the system is very close to smooth pipe behavior and the value is close to the standard roughness values for copper tubing which is 0.0015 mm, and therefore, the calculated roughness is credible data [9].

Table 6: This table shows the uncertainty of the measured parameters from the lab. The pressure, flow rate, and temperature uncertainties were calculated and recorded for each run. In the flow rate and temperature uncertainties, the uncertainty of the flowmeter and thermocouple were accounted for prior to performing statistical analysis. The average of these uncertainties was used to reflect the uncertainty of each measured parameter. Calipers were used to measure the internal diameter of the small pipe and large pipe. The uncertainty was accepted as half of the resolution of the calipers. Similarly, the small pipe length and large pipe length uncertainties were concluded to be half of the resolution of the provided value in the lab document [8]. The uncertainty of the Reynolds number, friction factor, and loss coefficient were calculated through RSS propagations, where individual uncertainties were combined into one.

Parameter	Uncertainty	Description / Reference
Pressure (±psi)	0.001938	Average from run 3 (statistical)
Flow Rate (±LPM)	0.7312	Average from run 3 (stat. + res.)
Small Pipe Diameter (±mm)	0.0005	Half of caliper resolution
Large Pipe Diameter (±mm)	0.0005	Half of caliper resolution
Small Pipe Length (±mm)	0.05	[8]
Large Pipe Length (±mm)	0.05	[8]
Temperature (±ºC)	0.3071	Average from run 3 (stat. + res.)
Reynolds Number (±unitless)	2117.33	Average from run 3 (RSS)
Friction Factor (±unitless)	0.08225	Average from run 3 (RSS)
Loss Coefficient (±unitless)	0.0833	Average from run 3 elbow (RSS)

Entrance Length Sample Calculation

The entrance length is the distance that the fluid travels once it enters the pipe before achieving a fully developed profile. For both laminar and turbulent flow, the entrance length is dependent on the Reynolds number, *Re*, and the pipe diameter, *D*. All the runs in this experiment reflected a turbulent flow because the Reynolds number for each data set was greater than 2300. The entrance length $L_{entrance,turbulent}$, was then able to be calculated as shown by the example below for the third large pipe data set of run 3. The fact that all of the Reynolds Numbers are turbulent indicate that the entrance length is relatively small which essentially guarantees that flow will fully develop within the pipe.

$$L_{entrance,turbulent} = 4.4 * Re^{\frac{1}{6}} * D \quad Eqn. 8.1, p. 229 \text{ Fox \& McDonald [2]} \tag{15}$$

$$L_{entrance,turbulent} = 4.4 * 56770.06507^{\frac{1}{6}} * 18.604$$

$$\therefore L_{entrance,turbulent} = 507.47\ mm$$

Entrance Length Uncertainty Sample Calculation

The uncertainty in the entrance length was found through an RSS propagation. The partial derivatives of the equation are taken with respect to each variable and multiplied by the uncertainties of that variable. The Reynolds number uncertainty is known from (8) and the diameter uncertainty is also known as a constant equal to 0.0127 millimeters.

$$\mu_{L_{entrance,turbulent}} = \sqrt{(\mu_{Re}\frac{\delta L_{entrance,turbulent}}{\delta Re})^2 + (\mu_D\frac{\delta L_{entrance,turbulent}}{\delta D})^2} \tag{16}$$

$$\mu_{L_{entrance,turbulent}} = \sqrt{(\mu_{Re}\frac{\delta L_{entrance,turbulent}}{\delta Re})^2 + (\mu_D\frac{\delta L_{entrance,turbulent}}{\delta D})^2}$$

$$\frac{\delta L_{entrance,turbulent}}{\delta Re} = \frac{4.4D}{6}Re^{-\frac{5}{6}} = \frac{4.4 * 18.604}{6} * 56770.0651^{-\frac{5}{6}} = 0.00149\ mm$$

$$\frac{\delta L_{entrance,turbulent}}{\delta D} = 4.4 * Re^{\frac{1}{6}} = 4.4 * 56770.0651^{\frac{1}{6}} = 27.2776$$

$$\mu_{L_{entrance,turbulent}} = \sqrt{(3325.22 * 0.00149\ mm)^2 + (0.0127\ mm * 27.2776)^2}$$

$$\therefore \mu_{L_{entrance,turbulent}} = 4.97\ mm$$

Table 7: The table below shows the Reynolds number and entrance length of all data sets for the large and small pipe. For all runs, the entrance length decreases as the percentage of the maximum flow rate and Reynolds number decreases The reason for this is that there is lower velocity, weaker inertial forces and dominate viscous effects at lower flow rates which allows the boundary layers to grow and merge over a shorter distance thus contributing to a shorter entrance length. For the small pipe run, the entrance length at the maximum flowrate is much less than that for the large pipe because a decreased cross-sectional area decreases flow rate and thus the same reasoning applies.

	Flow %		Flow Type	Reynolds Number	Entrance Length (mm)
	100		Turbulent	71558.19	537.43± 5.23
	90		Turbulent	64563.53	518.47± 5.08
	80		Turbulent	56611.98	507.24±4.97
	70		Turbulent	50479.65	497.64±4.89
	60	**Run 1**	Turbulent	42863.84	484.26±4.75
	50		Turbulent	36182.47	470.77±4.63
	40		Turbulent	29372.64	454.69±4.48
	30		Turbulent	21820.07	432.71±4.27
	20		Turbulent	14273.58	403.16±3.95
	10		Turbulent	7153.44	359.32±3.58
	100		Turbulent	71401.76	527.24±5.23
	90		Turbulent	64593.18	518.51±5.08
	80		Turbulent	56703.01	507.37±4.97
	70		Turbulent	50352.27	497.43±4.89
Large Pipe	60	**Run 2**	Turbulent	42708.06	483.96±4.75
	50		Turbulent	36299.32	471.02±4.62
	40		Turbulent	29389.85	454.74±4.48
	30		Turbulent	21778.93	432.58±4.27
	20		Turbulent	14353.31	403.54±3.95
	10		Turbulent	7114.53	358.99±3.58
	100		Turbulent	71602.47	527.49±5.23
	90		Turbulent	64643.37	518.58±5.08
	80		Turbulent	56770.07	507.47±4.97
	70		Turbulent	50419.41	497.54±4.89
	60	**Run 3**	Turbulent	42802.09	484.14±4.75
	50		Turbulent	36314.52	471.06±4.63
	40		Turbulent	29749.17	455.66±4.47
	30		Turbulent	21811.88	432.69±4.27
	20		Turbulent	14361.625	403.58±3.95
	10		Turbulent	7132.39	359.14±3.58
	100		Turbulent	130085.79	289.84±2.88
	90		Turbulent	116805.44	284.63±2.81
	80		Turbulent	103441.18	278.97±2.77
	70		Turbulent	92087.02	273.62±2.71
Small Pipe	60	**Run 4**	Turbulent	78433.78	266.39±2.63
	50		Turbulent	66451.33	259.14±2.56
	40		Turbulent	52492.42	249.15±2.47
	30		Turbulent	39745.46	237.86±2.36
	20		Turbulent	26075.29	221.73±2.21
	10		Turbulent	13216.62	197.99±1.96

Large Pipe Results

The large pipe results portray consistent trends along all 3 runs. The entry length decreased at a steady rate with the Reynolds number. This agrees with the turbulent flow theory and remains consistent with a fully developed flow behavior. The differences in entry lengths of different runs with the same flow percentages are very close together, which indicates strong repeatability in the data. Overall, the difference between the three runs is related to the head loss, seen through the standard deviation and total uncertainty.

Using 2 samples per iteration with a 5 second run time, run 1 showed the largest standard deviations and total uncertainties of head loss. This is due to the limitations of statistical analysis on a small sample size per unit time, where measurements were influenced significantly by any imperfections in the data.

With 200 samples per iteration and the same 5 second run time as run 1, run 2 showed a significant decrease in the standard deviations and total uncertainties of head loss when compared to run 1. With a larger sample size per unit time, the imperfections in readings were averaged out more easily. This led to measurements of head loss that were repeatable.

Run 3 used the same number of samples per iteration as run 2 with a run time that was twice as long, 200 samples per iteration and 10 seconds, respectively. While the number of samples per unit time remained the same as run 2, the longer run time led to a larger number of total samples. In comparison to run 2, the uncertainties were similar. While some uncertainties were less than those of run 2, they were often only marginally so. This suggests that the increased number of samples per iteration had a greater impact on the standard deviations and total uncertainties of head loss than the run time did.

Comparing all of the runs, run 3 seems to be the most accurate because it shows the most consistent Reynolds numbers and entrance lengths throughout all the flow percentages. As such, it was proven that while sampling stability does indeed improve measurement reliability, the sampling rate does not really affect these trends.

Table 8: This table shows the nine valves identified in the lab as well as the associated loss coefficients. The nine valves were identified through examination of labels, markings, and physical characteristics such as handle design, stem type, and body geometry. Through research on manufacturer catalogues and industry suppliers, the flow coefficient, C_v, was identified for each valve. Derived from substitutions for pressure change (17) and velocity (18) in the Darcy-Weisbach equation (19), the loss coefficient, K_L, was calculated based on the relationship between the internal diameter, d, and the flow coefficient (20).

$$C_v = Q\sqrt{\frac{SG}{\Delta P}} \text{ p. 2760 Oberg [10]} \tag{17}$$

$$v = \frac{Q}{A} = \frac{4Q}{\pi d^2} \tag{18}$$

$$\Delta P = K_L \frac{\rho v^2}{2} \tag{19}$$

$$K_L \cong \frac{891 \times d^4}{C_v^2} \tag{20}$$

Valve	Image	Description	Loss Coefficient
1		1” PVC True Union Ball Check Valve with FPM O-Rings [11]	$K_L = 7.36$
2		1” Milwaukee Valve One-Piece Butterfly Bronze Ball Valve [12]	$K_L = 0.48$
3		1” Rubinetterie Bresciane Brass Ball Valve [13]	$K_L = 0.36$
4		3/4” Rubinetterie Bresciane Three Way Ball Valve [14]	$K_L = 5.67$
5		1/2” Approved Vendor Piston Spring Check Valve [15]	$K_L = 5.79$

6			1” NIBCO Globe Valve [16]	$K_L = 7.23$
7			1” Milwaukee Valve Bronze Gate Valve [17]	$K_L = 0.31$
8			1/2” NOSHOK 504-MFC Needle Valve [18]	$K_L = 96.41$
9			3/4” Bray 3-Piece Ball Valve (Standard Port) [19]	$K_L = 1.96$

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Table 9: Lab Report Participation Log

Name	Date	Hours Worked	Description of Tasks Performed
Bryce Addeo	1/28/2026	1	Formatting first 8 pages of the report
Bryce Addeo	1/31/2026	2	Setting up calculations for table 1-4 values
Matthew Strogach	2/1/2026	2	Formatting pages 9-13 of the report/inputting data into tables 1-4
Sean Muller	2/1/2026	1	Data analysis for table 3
Bryce Addeo	2/1/2026	2	Data calculations for table 1, 2 and 4
Matthew Strogach	2/1/2026	1	Reynold's Number & Friction Factor graph
Bryce Addeo	2/1/2026	1	Flow Rate & Head Loss graph
Sean Muller	2/2/2026	1	Data verification and entry for table 3
Bryce Addeo	2/2/2026	1.5	Cover letter progress and table 1 description
Bryce Addeo	2/3/2026	1	Descriptions for tables 2, 3 and 4
Bryce Addeo	2/3/2026	1.5	Fixing tables and graphs at office hours
Bryce Addeo	2/3/2026	2	Head loss and uncertainty of head loss sample calculations
Matthew Strogach	2/7/2026	2.5	Reynolds Number and Friction Factor Uncertainties
Matthew Strogach	2/7/2026	2	Loss Coefficient Calculation and Uncertainty
Matthew Strogach	2/7/2026	4	Reynolds Number vs Friction Factor graph description, elbow head loss vs. Flow rate squared graph and description, error bars for figure 3, table 5
Bryce Addeo	2/7/2026	3	Completed descriptions for tables 1-4 and figure 5, cover letter progress, formatting
Bryce Addeo	2/7/2026	3	Cover letter progress, formatting, correcting errors
Matthew Strogach	2/8/2026	3	Roughness calculations and explanations
Matthew Strogach	2/8/2026	3	Entrance Length Calculations, uncertainty, and table
Sean Muller	2/8/2026	5	Valve table and loss coefficient calculations
Sean Muller	2/8/2026	0.5	Table formatting
Bryce Addeo	2/8/2026	4	Description for table 5, description for table 7, finished cover letter, corrected errors, formatting
Matthew Strogach	2/8/2026	1	Editing, writing sources
Sean Muller	2/8/2026	3.5	Valve table description, uncertainty table, results
Bryce Addeo	2/8/2026	2.5	Proofreading, correcting errors, final formatting
	Total Hours:	54	

Total Hours per Group Member: Bryce: 24.5 hrs, Matt: 18.5 hrs, Sean: 11 hrs